

Technical Note: Automated CRM Client Sync

How it works

A Python script (sync.py) reads the incoming CSV, validates and cleans each row, and upserts records into a local SQLite database (crm.db) in one command. It handles eight data quality issues found in the sample: duplicate client IDs (keeps the most recently updated record), inconsistent date formats (normalizes to YYYY-MM-DD), comma-formatted numbers, invalid emails (flagged, not rejected), missing phone numbers, missing portfolio values, a negative portfolio value of -50,000 (flagged for review -- every other value in the file is between 640,000 and 2,100,000, so this is almost certainly a sign flip), and inconsistent UAE phone formatting (all normalized to +971 XX XXX XXXX). Every issue is logged to sync_log.txt and stored in a sync_issues table alongside a sync_runs audit row for each run.

Keeping the data accurate and secure over time

The script validates column headers before processing any rows and aborts if something is missing, so a provider format change does not silently corrupt the database. For recurring use I would schedule it via cron and review the sync_issues table weekly for patterns. For security, crm.db should have restricted file permissions, the incoming CSV should be archived after each import rather than left in a shared folder, and credentials should live in environment variables if this ever moves to a server.

What is most likely to go wrong

A renamed column that still has the right field count would pass the header check and load incorrect data into the wrong column. The fix is matching exact column names rather than just checking for missing ones. The other risk is an interrupted run leaving the database in a partial state, but the entire import is already wrapped in a single transaction, so a mid-run crash rolls everything back cleanly.